

PORT EN BESSIN, France

WMO: 070290

Lat: 49.350N

Lon: 0.767W

Elev: 240

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 86-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.8	30.2	19.4	14.7	28.7	23.6	18.0	32.6	39.0	44.2	35.4	44.2	9.0	120	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.0	75.3	66.2	71.8	65.0	69.5	63.8	68.0	72.8	66.2	70.4	64.8	68.4	10.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	98.1	70.5	64.5	91.7	68.2	63.2	87.6	66.9	32.4	73.2	31.0	70.6	29.9	68.6	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.5	29.5	26.2	DB	22.4	83.4	7.1	4.8	17.3	86.8	13.2	89.7	9.2	92.4	4.0	95.9	(4)
(5)			WB	20.9	70.7	6.6	2.3	16.2	72.4	12.4	73.8	8.7	75.1	3.9	76.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	42.6	42.9	45.8	49.0	53.8	58.3	61.6	62.6	60.9	55.4	48.2	44.1		
	DBStd	8.77	6.05	6.58	5.53	4.66	4.11	4.37	3.37	3.81	4.36	4.64	4.71	5.76		
	HDD50	968	234	206	152	73	9	0	0	0	0	10	88	196		
	HDD65	4743	695	618	595	481	349	207	116	94	137	297	505	649		
	CDD50	1754	4	9	23	42	126	250	361	391	326	178	32	12		
	CDD65	52	0	0	0	0	1	7	11	19	13	1	0	0		
	CDH74	183	0	0	0	0	16	31	22	73	38	3	0	0		
	CDH80	38	0	0	0	0	4	7	2	18	7	0	0	0		
(14) Wind	WSAvg	13.1	15.2	16.0	14.2	13.6	11.9	10.9	10.9	10.9	11.7	12.8	13.9	15.0		
(15) Precipitation	PrecAvg	33.3	3.3	2.7	2.2	2.4	2.2	2.2	2.4	2.4	2.4	3.7	3.9	3.8		
	PrecMax	43.6	7.7	5.5	6.7	5.8	5.0	5.5	4.7	5.4	4.7	7.1	7.6	9.0		
	PrecMin	24.4	0.6	0.5	0.6	0.6	0.7	0.6	0.6	0.6	0.3	0.5	1.0	0.7		
	PrecStd	4.9	1.7	1.5	1.4	1.3	1.1	1.1	1.1	1.5	1.2	1.6	1.6	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	59.0	64.1	66.6	73.9	77.7	77.3	82.9	79.0	71.6	61.8	58.0		
		MCWB	53.0	52.9	54.8	57.1	61.5	67.2	68.3	67.1	67.0	62.3	58.3	55.5		
	2%	DB	54.1	56.0	58.9	61.6	66.4	71.2	72.7	75.1	72.1	66.5	58.4	55.8		
		MCWB	51.7	51.7	53.3	54.7	60.1	64.0	65.8	67.1	64.9	60.7	55.2	53.6		
	5%	DB	52.4	53.9	56.1	58.4	62.8	67.9	70.0	71.3	69.3	63.8	56.6	54.0		
		MCWB	50.0	50.8	52.2	53.1	57.7	62.3	64.6	65.4	63.8	59.3	53.8	51.9		
	10%	DB	50.6	51.7	53.7	55.8	60.2	65.0	67.8	68.6	67.2	61.8	55.0	52.2		
		MCWB	48.5	49.3	50.6	51.9	55.8	60.5	63.2	63.5	62.6	58.2	52.6	50.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	54.6	56.7	58.6	64.4	69.3	69.7	70.5	69.1	63.8	59.2	56.2		
		MCDB	55.0	57.4	62.5	63.7	70.5	75.5	74.3	76.6	73.5	68.6	60.5	57.4		
	2%	WB	52.1	52.8	54.2	55.8	60.5	65.0	67.1	68.2	66.4	61.6	55.8	54.0		
		MCDB	53.7	55.1	57.7	60.0	65.2	69.7	71.3	73.0	70.5	64.9	57.3	55.6		
	5%	WB	50.4	51.1	52.7	53.8	58.2	62.8	65.3	66.1	64.8	60.2	54.4	52.2		
		MCDB	52.0	53.4	55.6	57.5	62.0	66.7	69.1	70.1	68.0	63.0	56.1	53.7		
	10%	WB	48.7	49.4	50.9	52.1	56.4	61.1	63.7	64.7	63.5	58.8	52.9	50.4		
		MCDB	50.3	51.8	53.5	55.2	59.7	64.7	67.0	68.3	66.5	61.3	54.8	52.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	7.6	9.1	9.5	9.6	10.5	10.2	11.0	10.5	9.6	8.5	7.4		
		MCDBR	8.0	10.2	12.9	13.2	14.6	15.4	13.6	15.2	13.5	12.2	9.4	7.9		
		MCWBR	7.5	7.6	8.8	8.2	9.2	9.2	8.0	8.7	8.0	7.8	7.7	7.2		
	5% WB	MCDBR	7.7	8.3	12.0	11.8	13.6	13.7	12.3	12.9	12.1	11.2	8.6	8.0		
		MCWBR	7.8	7.3	8.8	7.9	9.4	9.1	8.0	8.7	7.7	7.9	7.8	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.346	0.379	0.395	0.401	0.398	0.392	0.386	0.373	0.359	0.335	0.323		
	taud		2.425	2.399	2.305	2.282	2.310	2.350	2.369	2.406	2.426	2.453	2.448	2.393		
	Ebn at Noon		228	252	261	270	272	274	273	270	262	244	224	211		
	Edh at Noon		21	26	34	39	39	38	36	34	30	25	20	19		
(44) All-Sky Solar Radiation	RadAvg		285	518	894	1403	1656	1843	1776	1479	1138	642	347	233		
	RadStd		26	62	96	125	129	181	115	103	74	53	42	29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page